
MACHINE LEARNING LECTURE

K-Means Clustering

Understanding clustering, centroid optimization, the elbow method, and silhouette analysis for selecting the optimal number of clusters.

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Unsupervised Learning

Introduction to K-Means

UNSUPERVISED CLUSTERING

K-Means is an unsupervised machine learning algorithm used to divide data into multiple clusters based on similarity. Instead of using labeled outputs, the algorithm discovers hidden structure directly from the dataset.

Goal: Minimize distance between data points and cluster centroids.

- No labeled outputs required
- Data partitioned into K groups
- Each cluster represented by a centroid
- Works using iterative optimization
- Groups similar points together
- Widely used in real-world ML applications

Applications include customer segmentation, recommendation systems, image compression, document clustering, and anomaly detection.

K-Means Clustering

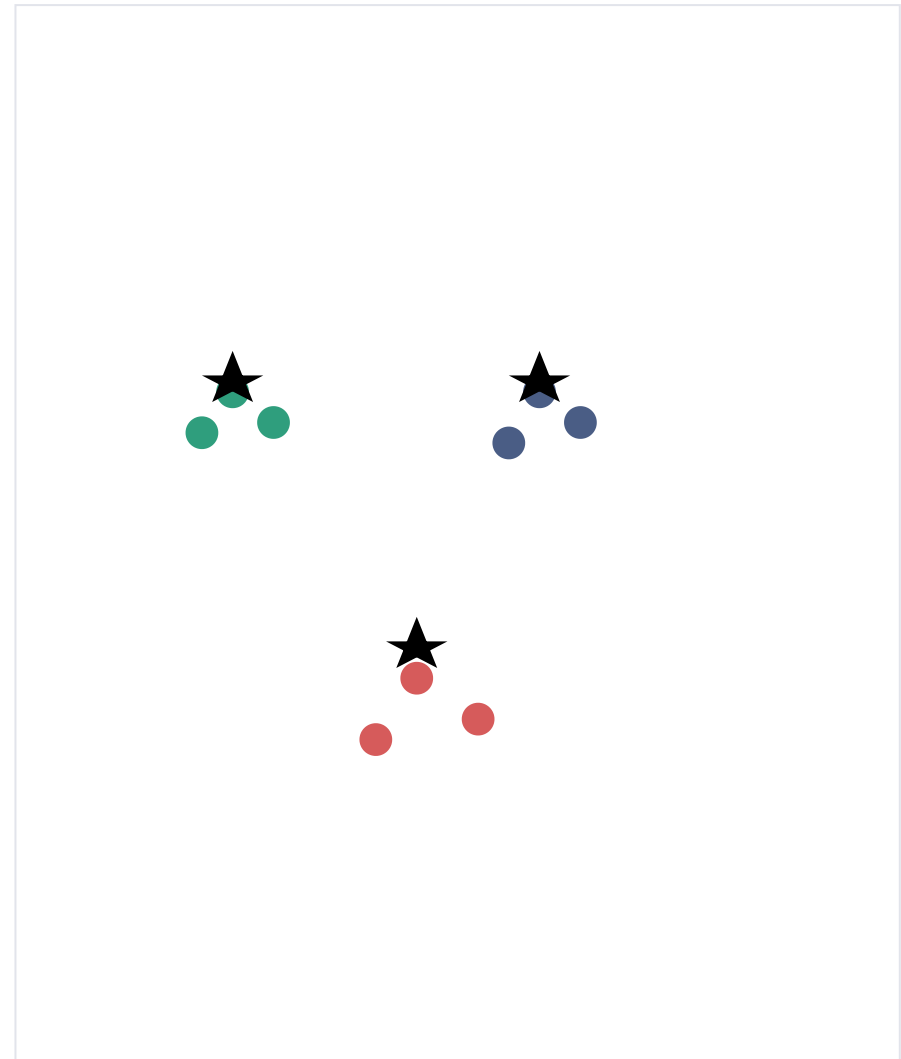
DISTANCE-BASED CLUSTERING

K-Means groups data points according to similarity using the distance from cluster centroids.

$$d = \sqrt{(x - c_x)^2 + (y - c_y)^2}$$

- $\mathbf{x} \rightarrow$ data point
- $\mathbf{c} \rightarrow$ centroid
- Each point assigned to nearest centroid
- Centroids updated iteratively

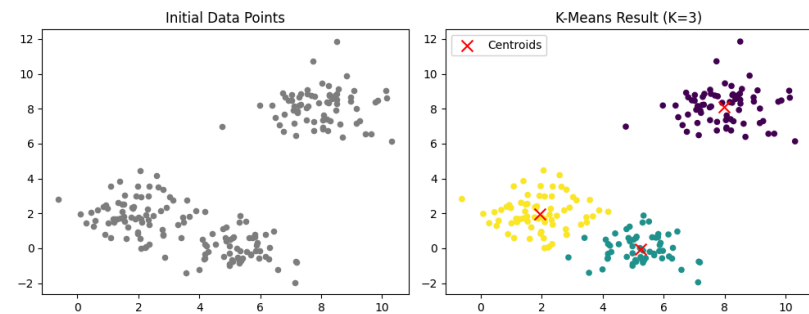
Goal: create compact clusters with minimum internal distance.



K-Means Algorithm

ITERATIVE OPTIMIZATION

- **Step 1:** Choose number of clusters K
- **Step 2:** Randomly initialize centroids
- **Step 3:** Assign points to nearest centroid
- **Step 4:** Update centroids using cluster means
- **Step 5:** Repeat until convergence



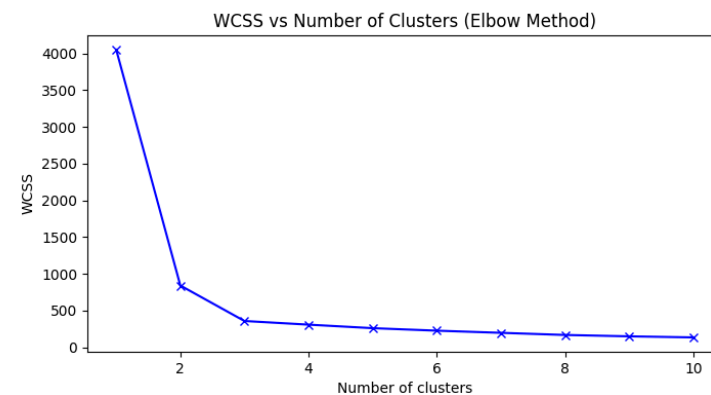
Choosing the Best K

ELBOW METHOD

Choosing K is extremely important because it directly affects cluster quality.

- Small K → underfitting
- Large K → overfitting
- No fixed universal value

Optimal K occurs where WCSS reduction slows.



Silhouette Score

Silhouette score measures how well a point fits within its assigned cluster compared to neighboring clusters.

$$s = (b - a) / \max(a, b)$$

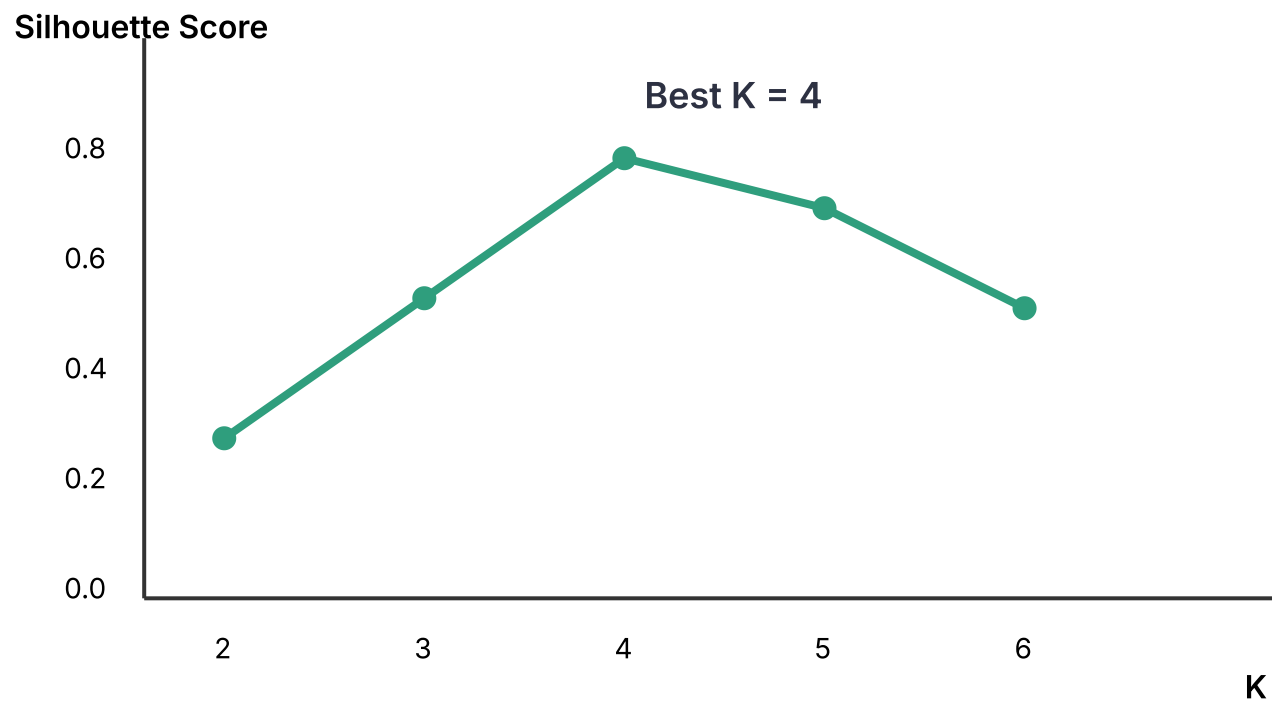
- **a** → intra-cluster distance
- **b** → nearest cluster distance
- Measures cohesion and separation simultaneously

- **s ≈ 1** → excellent clustering
- **s ≈ 0** → overlapping clusters
- **s < 0** → incorrect clustering

Higher silhouette score means better cluster quality.

Silhouette Analysis vs K

VALIDATION GRAPH



Advantages and Limitations

PRACTICAL UNDERSTANDING

Advantages

- Simple and easy to implement
- Computationally efficient on large datasets
- Works well for clearly separated clusters
- Fast convergence in most cases
- Easy to interpret and visualize

K-Means performs best when clusters are compact and spherical.

Limitations

- Requires predefined K value
- Sensitive to centroid initialization
- Affected heavily by outliers
- Struggles with irregular cluster shapes
- May converge to local minima

Different initial centroids can produce different clustering results.

Thank You

K-Means Clustering and Model Selection